Awami OT Kerning Mechanism

The shape of each glyph (here we use V[iolet], B[lue], G[reen], etc.) is represented conceptually by a sequence of vertical "slices"- rectangles of varying heights assumed to be 200 em-units wide. Together the slices serve as an approximation of the shape of the glyph, including marks. The glyph stream is enhanced to include marker glyphs indicating the tops and bottoms of the slices: `yt` = top and `yb` = bottom.

The kerning process involves progressively shifting the `yt` (top) markers of the leading glyph into the neighboring sequence of `yb` (bottom) markers of the trailing glyph. Each shift operation represents 200 units of kerning. As each slice is shifted, slices that have already been shifted must also be shifted again. The process works slice by slice until a collision occurs or the maximum number of shift operations (6 = 1200 em-units) have been performed. A collision is indicated by the bottom of the trailing-glyph slice being lower than the top of the leading-glyph slice (well, sort of, keep reading). The `ytk` markers indicate the tops of slices that have already been shifted.

The ultimate goal of the process is to calculate the amount of desired kerning which is indicated by the `kw` marker located in the stream at the point of kerning (the break between contextual glyph sequences). At each step, the `kw` value gets incemented by 200. Notice that when an internal collision occurs (Cases 2 and 3), the `kw` marker has already been incremented - erroneously as it turns out - and so must be decremented. Also at the end we subtract 100 units from the final result to avoid overly tight kerning. Both of these operations happen during the clean-up stage.

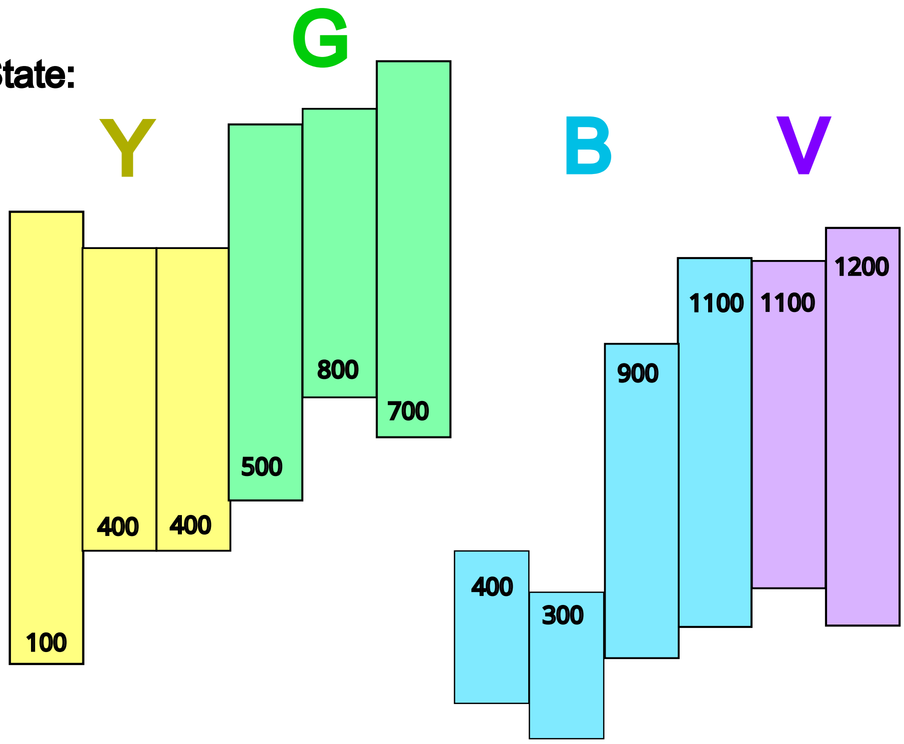
The marker sequence is simplified for these diagrams. Other markers are present that are used for calculations but not as part of the actual kerning algorithm. (The OT code uses the mark filtering mechanism to ignore all but the relevant types of markers.) Notice that in Case 3 we make a point to show Step 2a which involves shifting a slice from the following glyph. This appears fairly trivial in the diagram, but in actuality there are many intervening markers. Case 4 shows a more complicated example.

A simplication is that in these diagrams, Case 3 shows a top at 300 fitting under a bottom at 400. The implementation actually requires a difference of more than 200 units, so `yt300` would not fit under a bottom lower than `yb600`. Similar situations are present in Case 4.

Notice that we show the marker sequence from left to right although the glyph sequence flows from right to left.

CASE 1: Right-edge collision

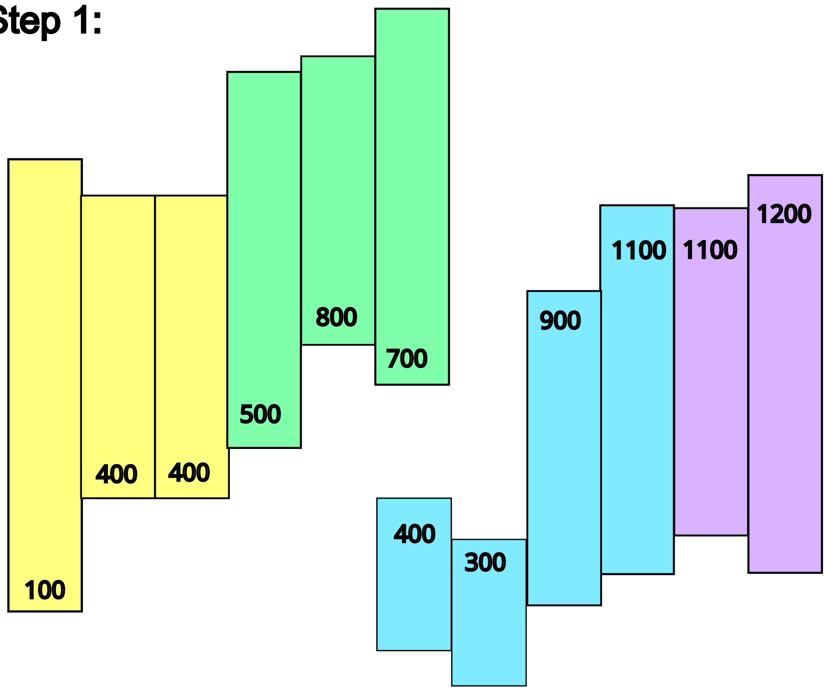
Initial State:



V yt1200 yt1100 B yt1100 yt900 yt300 yt400 kw0 G yb700 yb700 yb300 Y yb400 yb400 yb100

Step 1:

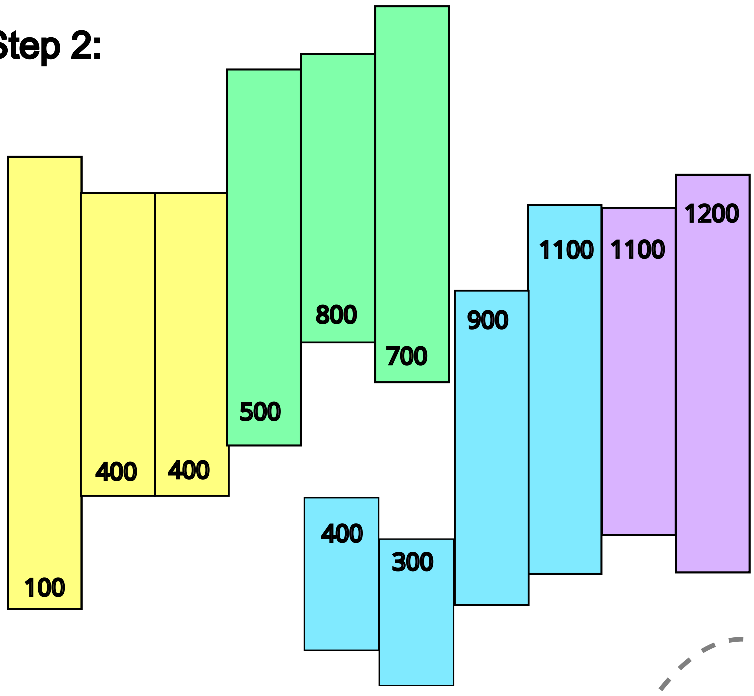
After Step 1:



V yt1200 yt1100 B yt1100 yt900 yt300 kw200 G yb700 ytk400 yb800 yb500 Y yb400 yb400 yb100

Step 2:

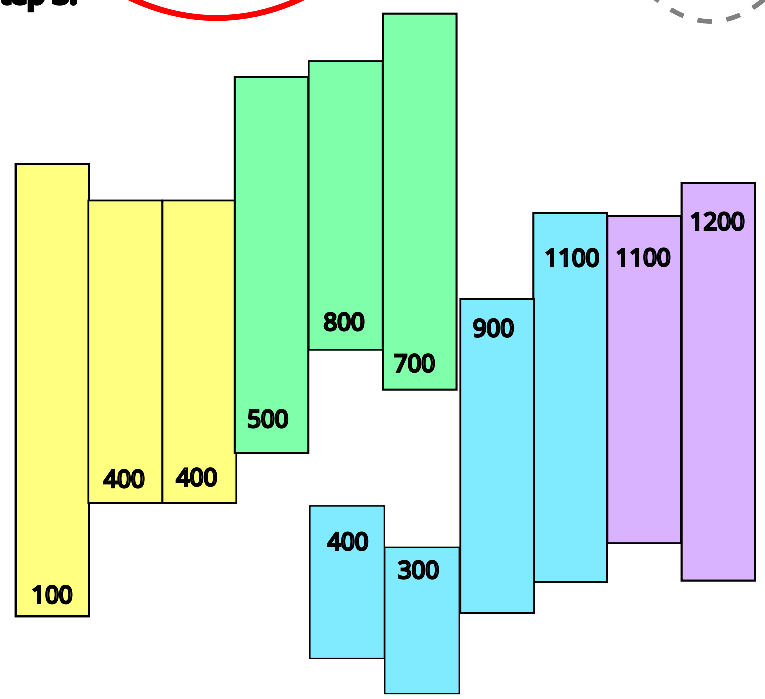
After Step 2:



V yt1200 yt1100 B yt1100 yt900 kw400 G yb700 ytk300 yb800 ytk400 yb500 Y yb400 yb400...

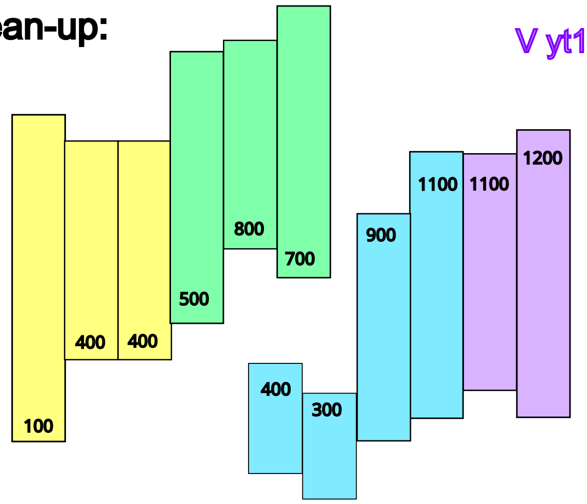
Step 3:

After Step 3:



V yt1200 yt1100 B yt1100 yt900 kblockR kw400 G yb700 ytk300 yb800 ytk400 yb500 Y yb400...

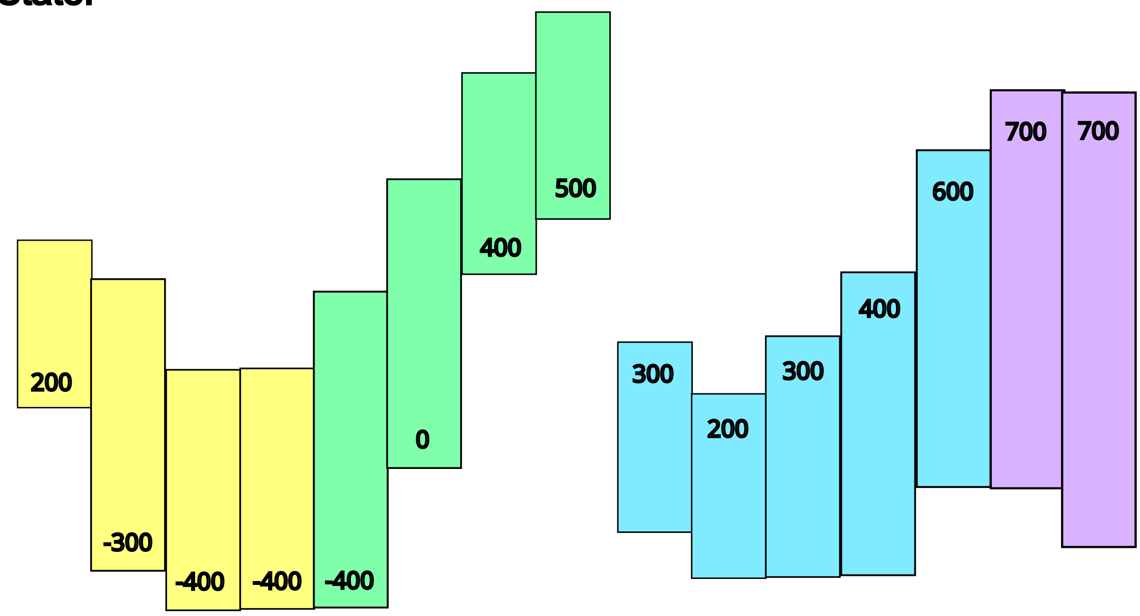
After Clean-up:



V yt1200 yt1100 B yt1100 yt900 kblockR kw300
G yb700 ytk300 yb800 ytk300 yb500...
Y yb400 yb400 yb100

CASE 2: Internal collision

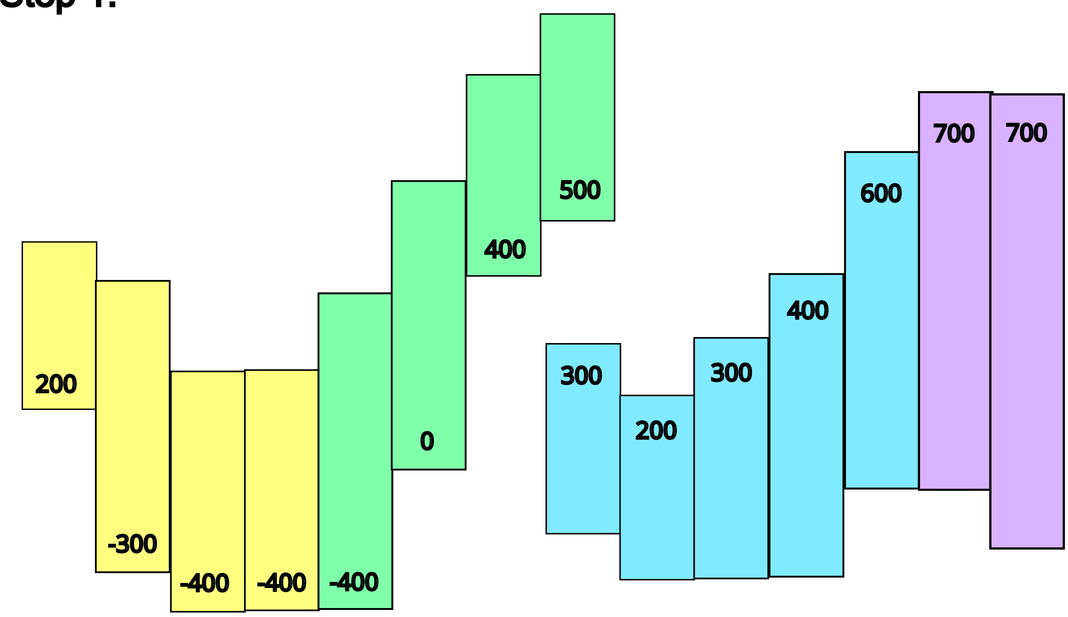
Initial State:



...yt700 B yt600 yt400 yt300 yt200 yt300 kw0 G yb500 yb400 yb0 ybN400 Y ybN400...

Step 1:

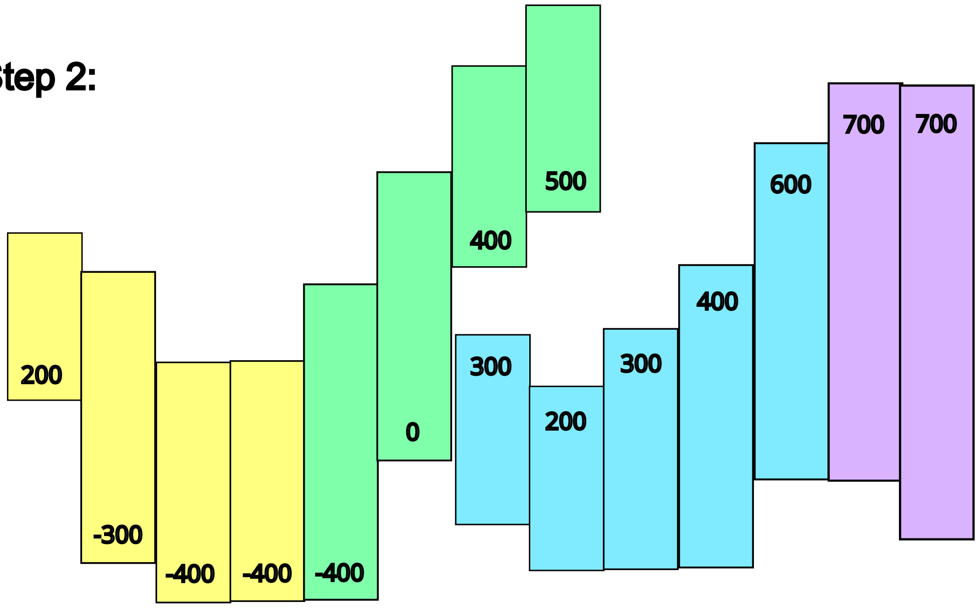
After Step 1:



...yt700 B yt600 yt400 yt300 yt200 kw200 G yb500 ytk300 yb400 yb0 ybN400 Y ybN400...

Step 2:

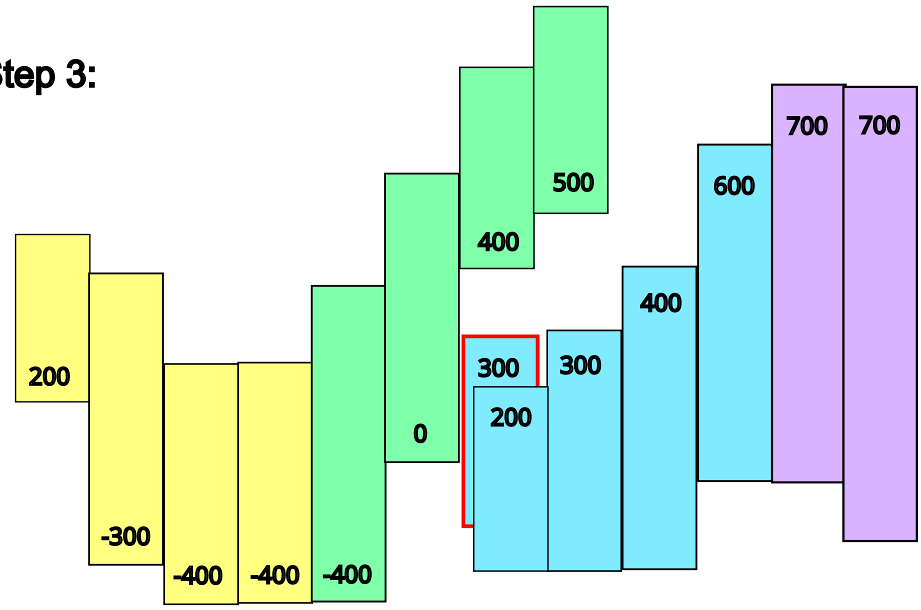
After Step 2:



...yt700 B yt600 yt400 yt300 kw400 G yb500 ytk200 yb400 ytk300 yb0 ybN400 Y ybN400...

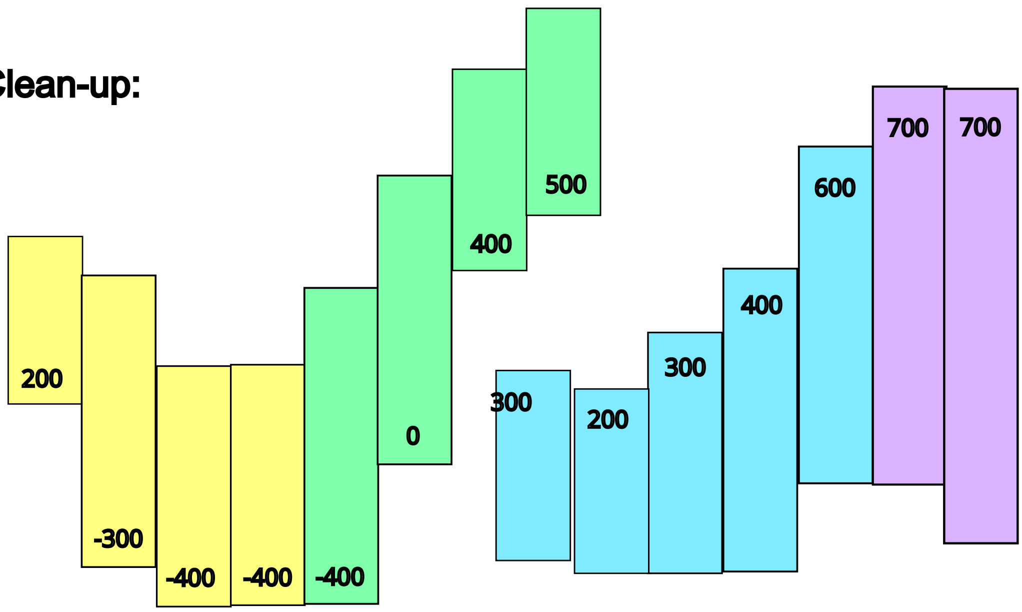
Step 3:

After Step 3:



...yt700 B yt600 yt400 yt300 kw600 G yb500 ytk300 yb400 ytk200 kblockl yb0 ybN400 Y...

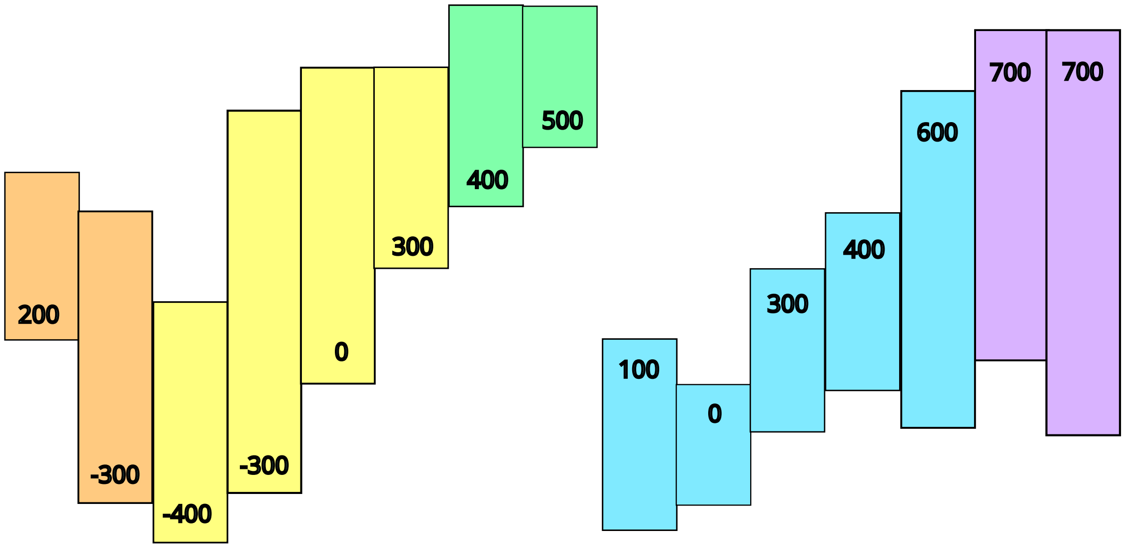
After Clean-up:



...B yt600 yt400 yt300 kblock kw300 G yb500 ytk300 yb400 ytk200 kblockl yb0 ybN400 Y...

CASE 3: Internal collision with second glyph in trailing sequence

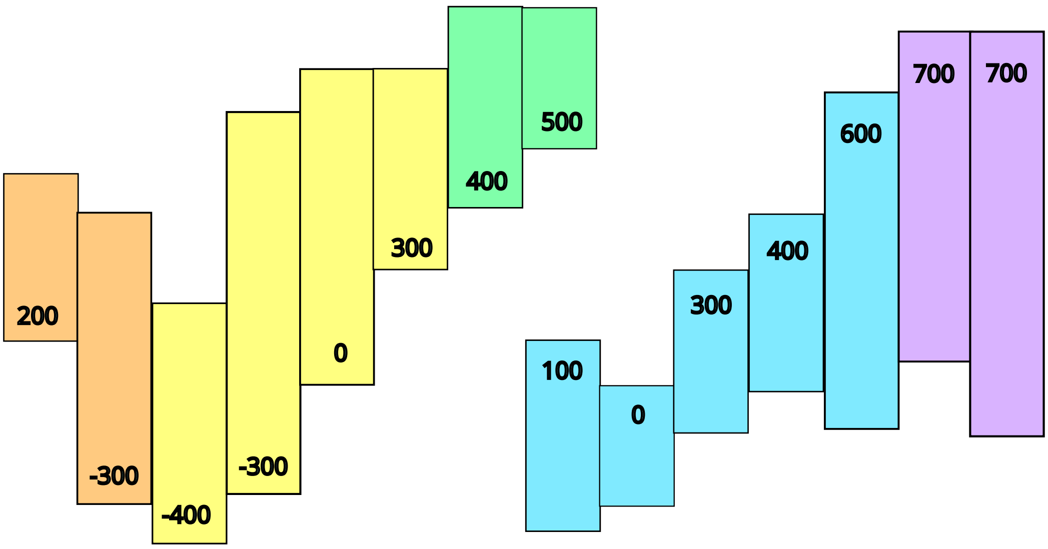
Initial State:



...yt700 B yt600 yt400 yt300 yt0 yt100 kw0 G yb500 yb400 Y b300 yb0 ybN300...

Step 1:

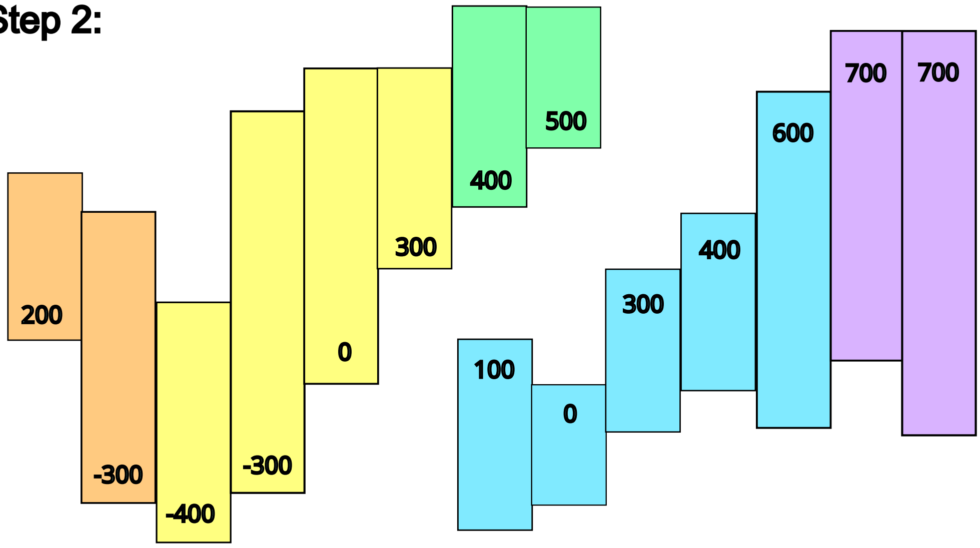
After Step 1:



...yt700 B yt600 yt400 yt300 yt0 kw200 G yb500 ytk100 yb400 Y yb300 yb0 ybN300...

Step 2:

After Step 2:



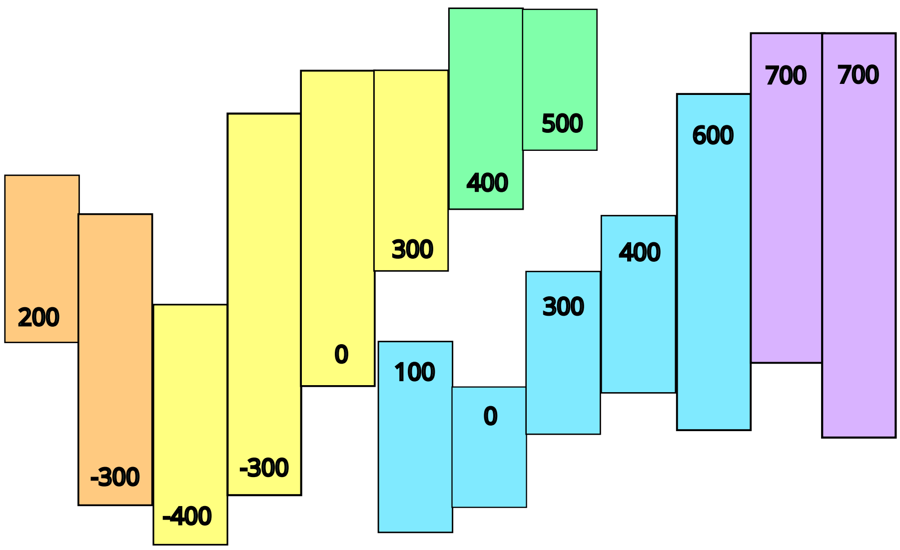
...yt700 B yt600 yt400 yt300 kw400 G yb500 ytk0 yb400 ytk100 Y yb300 yb0 ybN300...

After Step 2a: shift from next glyph

...yt700 yB yt600 yt400 yt300 kw400 G yb500 ytk0 yb400 ytk100 yb300 Y yb0 ybN300...

Step 3:

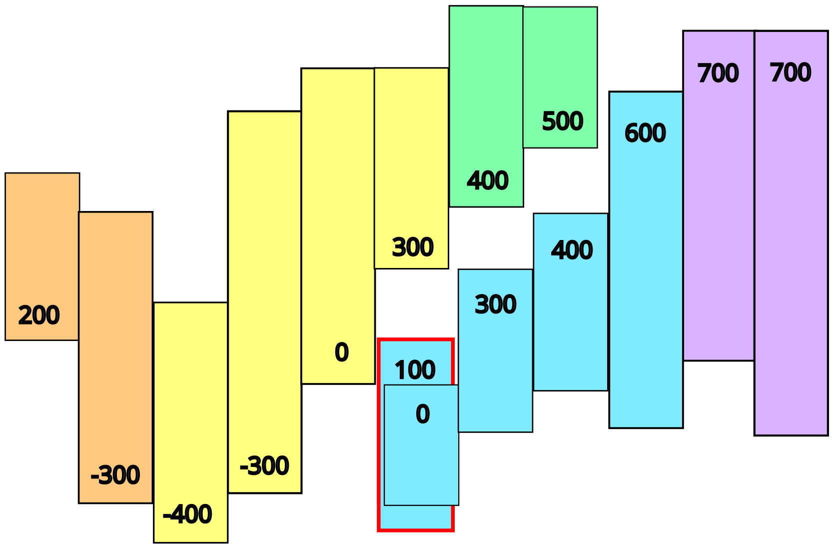
After Steps 3 and 3a:



...yt700 B yt600 yt400 kw600 G yb500 ytk300 yb400 ytk0 yb300 ytk100 yb0 Y ybN300 ybN400...

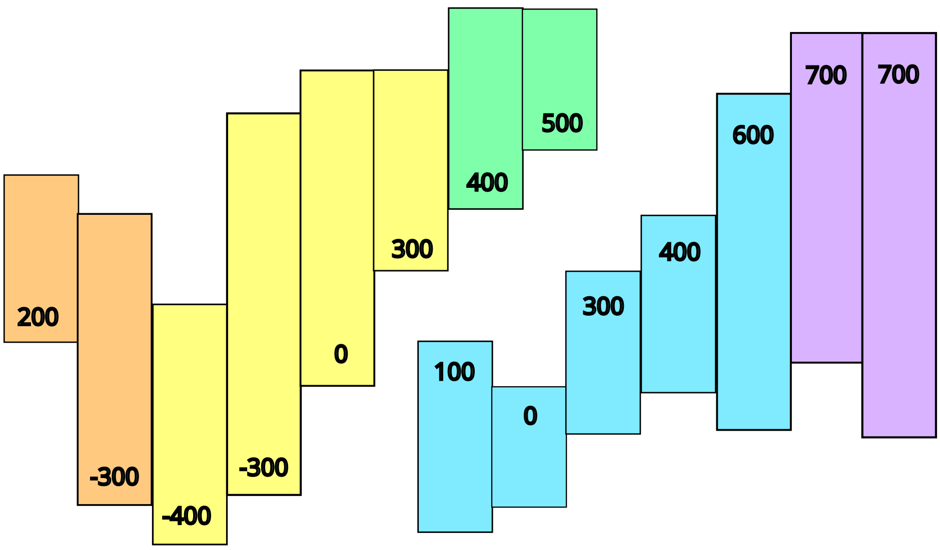
Step 4:

After Step 4:



...B yt600 kw800 G yb500 ytk400 yb400 ytk300 yb300 ytk0 kblockl yb0 Y ybN300 ybN400...

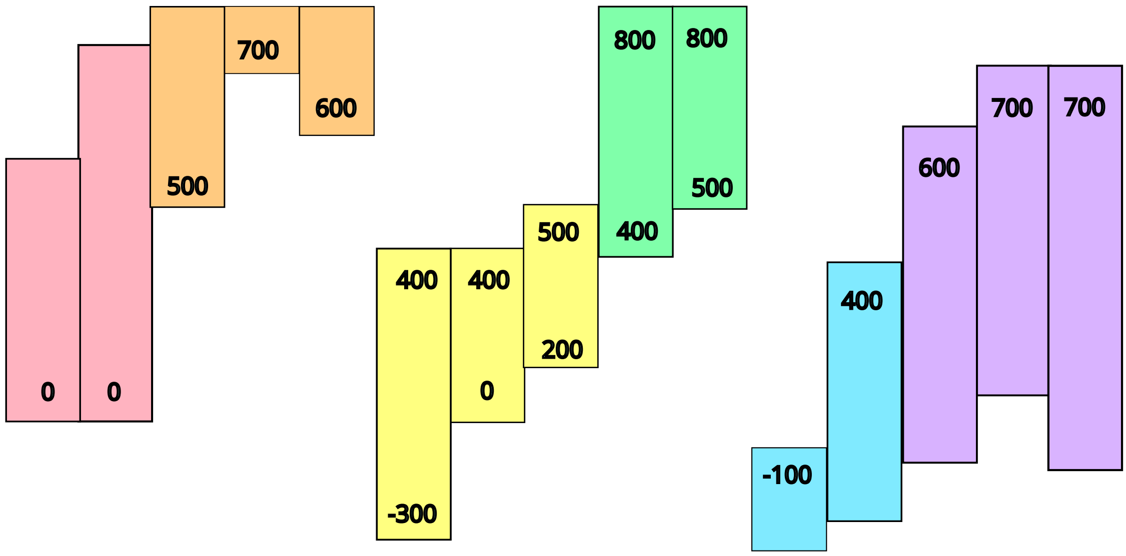
After Clean-up:



...B yt600 kblock kw500 G b500 ytk400 yb400 ytk400 yb300 ytk0 kBlockl yb0 Y ybN300...

CASE 4: Kerning before and after short sequence

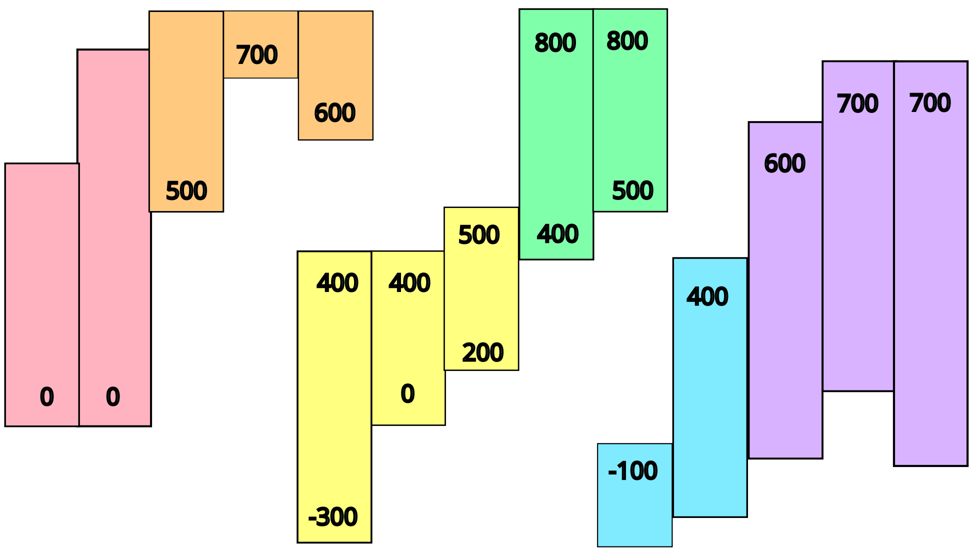
Initial State:



Step 1:

V yt700 yt700 yt600 B yt400 ytN100 kw0 G yb500 yb400 ybLast yt800 yt800
Y b200 yb0 ybN300 ybLast yt500 yt400 yt400 kw0 O b600 yb700 yb500 P yb0 yb0

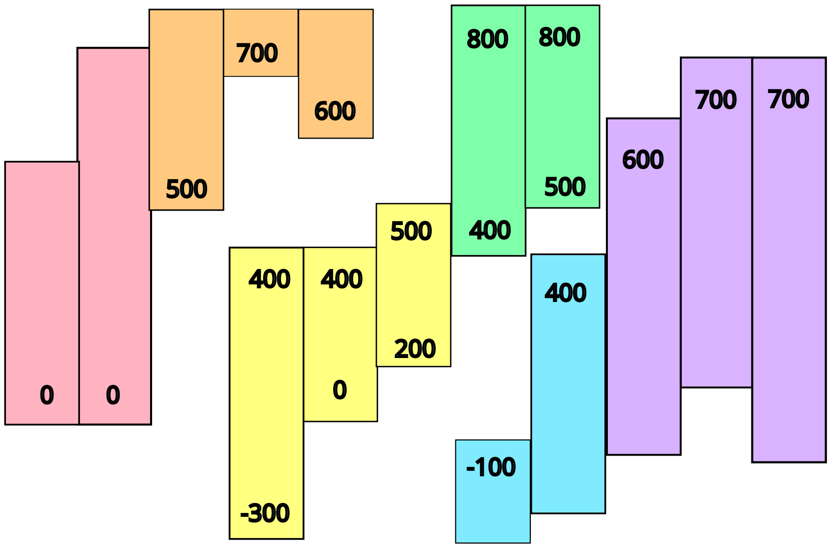
After Step 1:



Step 2:

V yt700 yt700 yt600 B yt400 kw200 G yb500 ytkN100 yb400 ybLast yt800 yt800...
Y b200 yb0 ybN300 ybLast yt500 yt400 kw200 O b600 ytk400 yb700 yb500 P yb0 yb0

After Step 2:



V yt700 yt700 yt600 B kw400 G yb500 ytk400 yb400 ytkN100 ybLast yt800 yt800
Y yb200 yb0 ybN300 ybLast yt500 kw400 O yb600 ytk400 yb700 ytk400 yb500 P yb0 yb0

After Step 2a: shift from next glyph

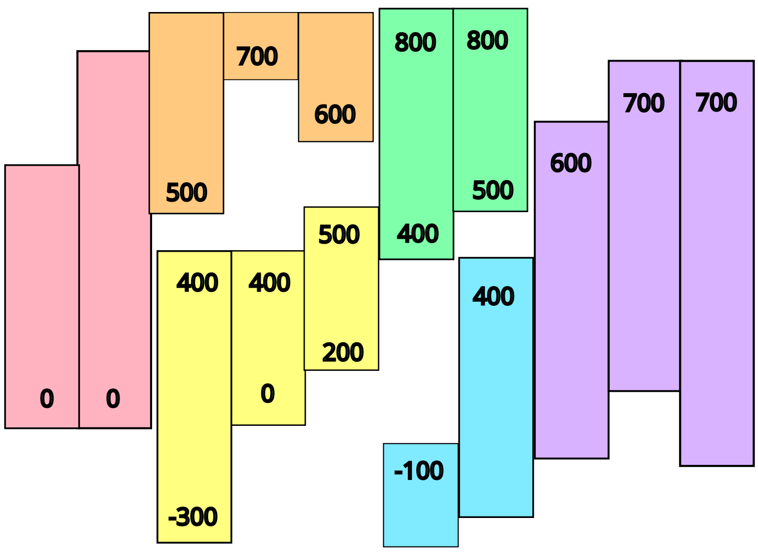
V yt700 yt700 yt600 B kw400 G yb500 ytk400 yb400 ytkN100 yb200 ybLast yt800 yt800
Y yb0 ybN300 ybLast yt500 kw400 O yb600 ytk400 yb700 ytk400 yb500 P b0 yb0

After Step 2b: shift from previous glyph

Step 3:

V yt700 yt700 B yt600 kw400 yb500 ytk400 yb400 ytkN100 yb200 ybLast yt800 yt800
Y b0 ybN300 ybLast yt500 kw400 O yb600 ytk400 yb700 ytk400 yb500 P yb0 yb0

After Step 3:

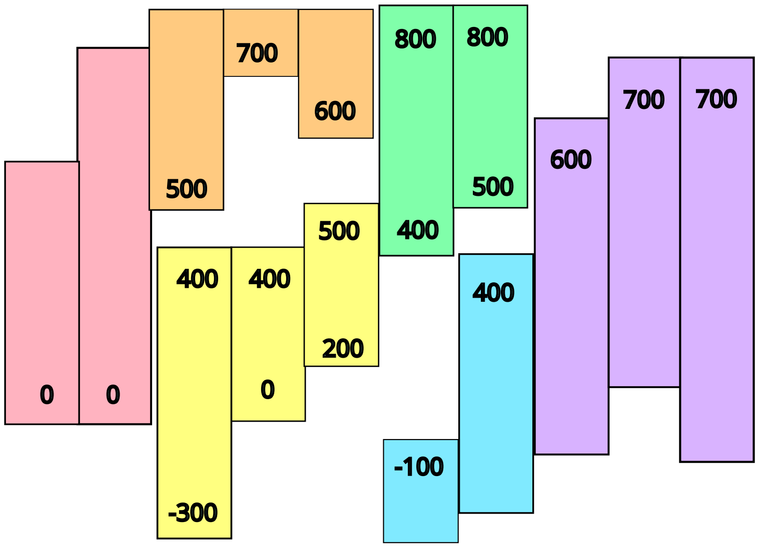


V yt700 yt700 B kblock yt600 kw400 G yb500 ytk400 yb400 ytkN100 yb200 ybLast yt800 yt800
Y yb0 ybN300 ybLast kw600 O yb600 ytk500 yb700 ytk400 yb500 ytk400 P yb0 yb0

After Step 3b: shift from previous glyph

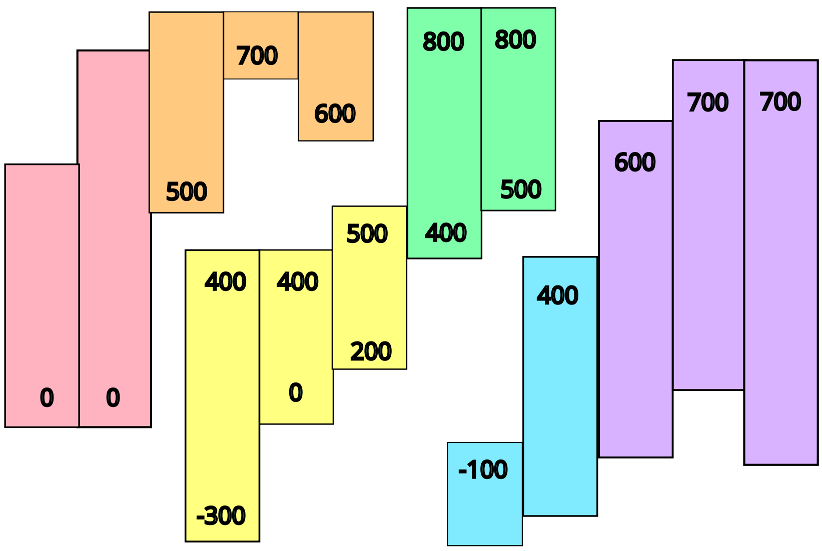
V yt700 yt700 B kblock yt600 kw400 G yb500 ytk400 yb400 ytkN100 yb200 ybLast yt800...
Y yb0 ybN300 ybLast yt800 kw600 O yb600 ytk500 yb700 ytk400 yb500 ytk400 P...
Step 4:

After Step 4:



V yt700 yt700 B kblockR yt600 kw400 yb500 ytk400 yb400 ytkN100 yb200 ybLast yt800
Y yb0 ybN300 ybLast yt800 kblockR kw600 O yb600 ytk500 yb700 ytk400 yb500 ytk400...

After Clean-up:



V yt700 yt700 B kblockR yt600 kw300 G yb500 ytk400 yb400 ytkN100 yb200 ybLast yt800
Y yb0 ybN300 ybLast yt800 kblockR kw500 O yb600 ytk500 yb700 ytk400 yb500 ytk400...